

A Direct Linear-Algebra Proof of the Carlen–Cordero-Erausquin Radial-Isotropic Criterion

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Status and Source

This packet answers the proof-style question in Carlen–Cordero-Erausquin, *Subadditivity of the Entropy and its Relation to Brascamp–Lieb Type Inequalities*, arXiv:0710.0870 [1]. On page 30, just before Theorem 4.3, the authors write that the theorem concerns a problem in linear algebra and that they do not know a direct proof in a purely linear algebra context. The theorem itself is already proved in the source via entropy and Brascamp–Lieb extremizers. The contribution here is a direct proof of the same linear-algebra criterion.

where Φ_A^* denotes the Legendre transform of Φ_A . Since $\nabla\Phi_A^*(\nabla\Phi_A(0)) = 0$, the choice $c = \nabla\Phi_A(0)$ minimizes $\Phi_A^*(c)$, and hence $D(A, c) + \sum_{j=1}^m c_j \ln(c_j)$. There is a misprint in [11] in which it is stated (in slightly different notation) that this choice of c minimizes $D(A, c)$ itself.

We finally return to Lemma 3.5, as we are now in a position to give necessary and sufficient conditions for the existence of the change of variables provided there.

Let $A = \{a_1, \dots, a_m\}$ be family of m vectors spanning $\mathbb{R}^n$, and let c be any vector in $\mathbb{R}^m$ with $0 \leq c_j \leq 1$ for all j . Theorem 3.9 gives us necessary and sufficient conditions for the existence of an extremal X for the subadditivity inequality. By Theorem 2.2, these conditions are also necessary and sufficient for the existence of extremals for the Brascamp–Lieb inequality. Moreover, we see that extremals for the latter exist if and only if centered Gaussian extremals exist.

From here, it is easy to prove the following theorem which supersedes Lemma 3.5, and gives necessary and sufficient conditions for the existence of the change of variables considered there. This result was obtained (in the more general multidimensional setting) by Bennett–Carbery–Christ–Tao [6] along their study of the Brascamp–Lieb extremizers ; here we use the extremizers to the subadditivity of entropy inequality. Though this theorem concerns a problem in linear algebra, we do not know a direct proof of it in a purely linear algebra context, though there may be one.

4.3 THEOREM. *Let $\{a_1, \dots, a_m\}$ be any collection of m vectors that span $\mathbb{R}^n$ for $n \geq 2$. Let $\{c_1, \dots, c_m\}$ be any m numbers satisfying $0 \leq c_j \leq 1$ for each j . Then there exists an invertible symmetric matrix $n \times n$ matrix R so that*

$$\sum_{j=1}^m c_j \left(\frac{Ra_j}{|Ra_j|} \right) \otimes \left(\frac{Ra_j}{|Ra_j|} \right) = \text{Id}_{\mathbb{R}^n}$$

if and only if the set $\{a_1, \dots, a_m\}$ is totally reducible for c

Source Question

Let $A = \{a_1, \dots, a_m\}$ span $\mathbb{R}^n$, $n \geq 2$, and let $0 \leq c_j \leq 1$. The source asks for a direct linear-algebra proof of the criterion for the existence of an invertible symmetric matrix R such that

$$\sum_{j=1}^m c_j \left(\frac{Ra_j}{|Ra_j|} \right) \otimes \left(\frac{Ra_j}{|Ra_j|} \right) = \text{Id}_{\mathbb{R}^n}. \quad (1)$$

The criterion is that A is totally reducible for c , in the sense of the source's Definition 3.8.

Definitions

For $J \subseteq \{1, \dots, m\}$, write

$$V_J = \text{span}\{a_j : j \in J\}.$$

The Brascamp–Lieb rank polytope associated with A is

$$K_A = \left\{ c \in [0, 1]^m : \sum_{j=1}^m c_j = n, \quad \sum_{j \in J} c_j \leq \dim V_J \text{ for all } J \right\}.$$

Its strict part is

$$K_A^\circ = \left\{ c \in K_A : \sum_{j \in J} c_j < \dim V_J \text{ for all nonempty proper } J \right\}.$$

After discarding zero-weight indices, A is *totally reducible for c* if the remaining index set can be partitioned into nonempty sets $J_1, \dots, J_r$ such that

$$\mathbb{R}^n = V_{J_1} \oplus \dots \oplus V_{J_r}, \quad c_{J_\ell} \in K_{A_{J_\ell}}^\circ \quad (\ell = 1, \dots, r),$$

where each A_{J_ℓ} is considered inside V_{J_ℓ} .

Proof Intuition

The equation is exactly the condition that a linear image of the vectors is in weighted radial isotropic position. For the strict/interior case, one can find the required change of variables by minimizing a log-det function. Cauchy–Binet rewrites this function as a log-sum-exp over the bases of the vector matroid of A . The rank inequalities defining K_A° are precisely the statement that c lies in the relative interior of the basis polytope, which makes the log-sum-exp coercive modulo its harmless lineality. At the minimum, the leverage-score equations give (1).

For boundary points, every saturated rank inequality forces an actual orthogonal splitting after any solution of (1): taking the trace against the orthogonal projection onto the saturated span leaves no room for positive weight vectors outside that span to have a component inside it. Conversely, if total reducibility gives a direct-sum decomposition into strict blocks, solve the strict problem on each block and assemble the block solutions. A final polar-decomposition step makes the resulting map symmetric positive definite.

Main Result

Theorem 1. *Let $A = \{a_1, \dots, a_m\}$ span $\mathbb{R}^n$, $n \geq 2$, and let $0 \leq c_j \leq 1$. Ignoring indices with $c_j = 0$, assume all positive-weight vectors are nonzero. Then there exists an invertible symmetric matrix R satisfying (1) if and only if A is totally reducible for c .*

Proof. We give a proof independent of entropy and Brascamp–Lieb extremizers.

Step 1: the strict case. Let $\mathcal{B}$ denote the family of n -element subsets $B \subseteq \{1, \dots, m\}$ for which $\{a_j : j \in B\}$ is a basis of $\mathbb{R}^n$. By the vector-matroid base-polytope theorem of Edmonds [2], the polytope

$$\text{conv}\{1_B : B \in \mathcal{B}\}$$

is exactly K_A . Thus $c \in K_A^\circ$ is the same as saying that c lies in the relative interior of this basis polytope.

Assume $c \in K_A^\circ$. For $t = (t_1, \dots, t_m) \in \mathbb{R}^m$, put

$$M(t) = \sum_{j=1}^m e^{t_j} a_j \otimes a_j$$

and define

$$F(t) = \log \det M(t) - \sum_{j=1}^m c_j t_j.$$

By Cauchy–Binet,

$$\det M(t) = \sum_{B \in \mathcal{B}} \det(A_B)^2 \exp \left(\sum_{j \in B} t_j \right),$$

where A_B is the $n \times n$ matrix with columns a_j , $j \in B$. Therefore

$$F(t) = \log \sum_{B \in \mathcal{B}} \det(A_B)^2 e^{\langle 1_B - c, t \rangle}.$$

Since $0 \in \text{relint conv}\{1_B - c : B \in \mathcal{B}\}$, this log-sum-exp is coercive on the quotient by its lineality space and hence attains its minimum. Let t^* be a minimizer. Differentiating at t^* gives, for every j ,

$$c_j = e^{t_j^*} \langle M(t^*)^{-1} a_j, a_j \rangle. \quad (2)$$

Set $R = M(t^*)^{-1/2}$, the positive square root. Then

$$\begin{aligned} \sum_{j=1}^m c_j \left(\frac{Ra_j}{|Ra_j|} \right) \otimes \left(\frac{Ra_j}{|Ra_j|} \right) &= \sum_{j=1}^m e^{t_j^*} Ra_j \otimes Ra_j \\ &= R \left(\sum_{j=1}^m e^{t_j^*} a_j \otimes a_j \right) R = \text{Id}. \end{aligned}$$

Thus the desired matrix exists in the strict case.

Step 2: a solution forces splitting on saturated sets. Suppose now that an invertible R satisfying (1) exists. Put

$$u_j = \frac{Ra_j}{|Ra_j|}$$

for all j with $c_j > 0$. Then

$$\sum_{j:c_j>0} c_j u_j \otimes u_j = \text{Id}. \quad (3)$$

Taking traces in (3) gives $\sum_j c_j = n$. If J is any set of positive weight indices and P_J is the orthogonal projection onto $\text{span}\{u_j : j \in J\}$, then

$$\dim \text{span}\{u_j : j \in J\} = \text{tr } P_J = \sum_i c_i |P_J u_i|^2 \geq \sum_{j \in J} c_j. \quad (4)$$

Because R is invertible, this is exactly the rank inequality for the original vectors a_j . Hence $c \in K_A$.

Moreover, equality in (4) for a nonempty proper J is rigid. Since $|P_J u_j| = 1$ for $j \in J$, equality can hold only if

$$P_J u_i = 0 \quad (i \notin J, c_i > 0). \quad (5)$$

Thus every positive-weight vector outside a saturated span is orthogonal to that span, after applying R .

If there is no nonempty proper saturated set among the positive-weight indices, then $c \in K_A^\circ$ and A is totally reducible with one block. Otherwise choose an inclusion-minimal nonempty saturated set J . Minimality implies $c_J \in K_{U_J}^\circ$, where $U_J = \{u_j : j \in J\}$ is considered in its span. By (5), the remaining positive-weight vectors lie in the orthogonal complement of $\text{span } U_J$, and (3) splits into two identity decompositions, one on $\text{span } U_J$ and one on its orthogonal complement. Repeating this finite procedure yields an orthogonal direct-sum decomposition for the vectors $\{u_j : c_j > 0\}$ into strict blocks. Applying R^{-1} transfers it to a direct-sum decomposition of the original spans V_{J_ℓ} . Therefore A is totally reducible for c .

Step 3: total reducibility gives a solution. Conversely, suppose A is totally reducible for c . Discard the indices with $c_j = 0$, and write the positive index set as a disjoint union

$$J_1 \cup \cdots \cup J_r$$

with

$$\mathbb{R}^n = V_{J_1} \oplus \cdots \oplus V_{J_r}, \quad c_{J_\ell} \in K_{A_{J_\ell}}^\circ.$$

Choose an invertible linear map T such that the subspaces

$$E_\ell := TV_{J_\ell}$$

are mutually orthogonal. The rank inequalities and their strictness are preserved by T , so

$$c_{J_\ell} \in K_{TA_{J_\ell}}^\circ$$

inside E_ℓ . By Step 1, for each ℓ there is a positive definite operator $R_\ell : E_\ell \rightarrow E_\ell$ such that

$$\sum_{j \in J_\ell} c_j \left(\frac{R_\ell T a_j}{|R_\ell T a_j|} \right) \otimes \left(\frac{R_\ell T a_j}{|R_\ell T a_j|} \right) = \text{Id}_{E_\ell}.$$

Let R_0 be the block-diagonal operator on the orthogonal sum $\mathbb{R}^n = E_1 \oplus \cdots \oplus E_r$ whose restriction to E_ℓ is R_ℓ . Then the invertible map $L = R_0 T$ satisfies

$$\sum_j c_j \left(\frac{L a_j}{|L a_j|} \right) \otimes \left(\frac{L a_j}{|L a_j|} \right) = \text{Id}.$$

Finally take the polar decomposition $L = QS$, where Q is orthogonal and S is symmetric positive definite. Since $La_j/|La_j| = Q(Sa_j/|Sa_j|)$, conjugating the last identity by Q^* gives

$$\sum_j c_j \left(\frac{Sa_j}{|Sa_j|} \right) \otimes \left(\frac{Sa_j}{|Sa_j|} \right) = \text{Id}.$$

Thus the required invertible symmetric matrix exists. This completes the direct proof. $\square$

Verification and Limitations

The proof is purely finite-dimensional linear algebra plus the standard vector matroid basis-polytope theorem. It does not use entropy, Fisher information, Brascamp–Lieb extremizers, or Gaussian optimization. The theorem being proved is not new: it is Theorem 4.3 of the source paper. The packet answers the authors’ request for a direct proof in a purely linear-algebra context.

Bounded novelty check: local run indexes had no ‘0710.0870’ packet. Web search for radial isotropic position, Tyler’s M-estimator, Forster transforms, and the normalized identity equation found close known formulations, especially the subspace-condition criterion for Tyler’s M-estimator and radial isotropic position. Therefore the novelty confidence is only for this compact direct derivation and its explicit connection to the source theorem, not for the criterion itself.

Human Review Recommendation

Review as a likely valid full answer to the source’s direct-proof request. The main items to check are: the identification of K_A with the vector matroid basis polytope, the existence of a minimizer of F in the relative interior case, and the transfer from an arbitrary invertible map to a symmetric positive map via polar decomposition.

References

- [1] E. A. Carlen and D. Cordero-Erausquin, *Subadditivity of the Entropy and its Relation to Brascamp–Lieb Type Inequalities*, arXiv:0710.0870, 2007.
- [2] J. Edmonds, *Submodular functions, matroids, and certain polyhedra*, in *Combinatorial Structures and their Applications*, Gordon and Breach, 1970, pp. 69–87.